MOHAMMED AASIK

ASPIRING MACHINE LEARNING ENGINEER

Erode, Tamil Nadu, India | Phone: 90255 31878 | Email: mdaasik.work@gmail.com
GitHub: github.com/mdaasik-tech | LinkedIn: linkedin.com/in/mdaasik-a-257787433

PROFESSIONAL SUMMARY

B.Sc. Artificial Intelligence and Machine Learning student with strong hands-on Python and machine learning project experience, including deployed anomaly detection applications. Built supervised, unsupervised, ensemble, and anomaly detection projects covering data preprocessing, exploratory data analysis, feature engineering, hyperparameter tuning, and model evaluation. Deployed ML applications using Streamlit and developing foundational backend skills with FastAPI. Seeking an entry-level role in AI/ML, Python development, data science, or backend engineering.

EDUCATION

B.Sc. Artificial Intelligence and Machine Learning
Hindusthan College of Science and Commerce, Erode

TECHNICAL SKILLS

Programming	Python, Object-Oriented Programming, Data Structures, Problem Solving
Machine Learning	Supervised Learning, Unsupervised Learning, Ensemble Learning, Anomaly Detection, Feature Engineering
Data Analysis & Visualization	NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis, Statistical Analysis
Data Preprocessing & Model Optimization	Data Cleaning, Missing Value Handling, Outlier Detection, Feature Scaling, Label Encoding, Train-Test Split, Cross Validation, Hyperparameter Tuning, GridSearchCV, RandomizedSearchCV, PCA Dimensionality Reduction
Evaluation Metrics	Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ROC-AUC, Silhouette Score
Backend Development	FastAPI, REST API Fundamentals, Pydantic, Uvicorn
Deployment	Streamlit, Machine Learning Application Deployment
Tools	Git, GitHub, Jupyter Notebook, Google Colab, VS Code

MACHINE LEARNING PROJECTS (ACADEMIC & PERSONAL)

MACHINE SENSOR ANOMALY DETECTION

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Isolation Forest, Streamlit

- Developed an unsupervised anomaly detection system to identify abnormal machine sensor behavior, designed around a predictive-maintenance use case.
- Cleaned data by handling infinite and missing values, and conducted exploratory data analysis to understand sensor patterns.
- Implemented Isolation Forest with hyperparameter tuning and analyzed anomaly scores and predictions to flag abnormal behavior.
- Designed and deployed an interactive Streamlit application.
- Live Demo: Streamlit application

NETWORK TRAFFIC ANOMALY DETECTION

Technologies: Python, Pandas, Scikit-learn, Isolation Forest, GridSearchCV

- Built an unsupervised machine learning system to detect unusual network traffic patterns.
- Performed preprocessing and exploratory analysis on network telemetry data.
- Implemented Isolation Forest and optimized hyperparameters with GridSearchCV; analyzed anomaly predictions and detection patterns.
- Designed the workflow around real-world network monitoring and security applications.

SERVER PERFORMANCE ANOMALY DETECTION

Technologies: Python, Pandas, Scikit-learn, One-Class SVM, Streamlit

- Developed a machine learning application to detect abnormal server performance behavior.
- Applied preprocessing techniques and implemented One-Class SVM to identify unusual observations.
- Analyzed model predictions to distinguish normal from anomalous server behavior.
- Built and deployed an interactive Streamlit application demonstrating the detection workflow.
- Live Demo: Streamlit application

USER BEHAVIOR ANOMALY DETECTION

Technologies: Python, Pandas, Scikit-learn, Isolation Forest, Streamlit

- Developed an anomaly detection application to identify unusual user behavior patterns.
- Performed data preprocessing and exploratory analysis; implemented Isolation Forest for unsupervised detection.
- Analyzed model predictions to identify potentially abnormal behavioral patterns.
- Built and deployed an interactive Streamlit application to demonstrate the detection workflow.
- Live Demo: Streamlit application

HUMAN ACTIVITY RECOGNITION USING PCA AND SVM

Technologies: Python, Pandas, NumPy, Scikit-learn, PCA, SVM

- Developed a dimensionality reduction and classification pipeline for human activity recognition.
- Preprocessed sensor-based activity data using feature scaling and label encoding.
- Applied PCA for dimensionality reduction and trained an SVM classifier on the transformed features.
- Evaluated classification performance using standard machine learning metrics.

CUSTOMER SEGMENTATION USING K-MEANS

Technologies: Python, Pandas, NumPy, Scikit-learn, K-Means, StandardScaler

- Developed an unsupervised customer segmentation system using K-Means clustering.
- Applied feature scaling before clustering and identified customer groups based on behavioral and spending patterns.
- Evaluated clustering quality using Silhouette Score and visualized customer segments for interpretation.

EMPLOYEE WORK PATTERN SEGMENTATION

Technologies: Python, Pandas, Scikit-learn, Hierarchical Clustering, SciPy, Matplotlib

- Built a hierarchical clustering model to identify employee work behavior patterns.
- Applied StandardScaler for feature normalization and generated dendrograms to analyze hierarchical relationships.
- Applied Agglomerative Clustering to create employee groups and evaluated cluster quality using Silhouette Score.

DRUGAWARBOT - AI AWARENESS CHATBOT

Technologies: Python, Streamlit, Generative AI, XGBoost

- Developed an AI-powered chatbot focused on drug abuse awareness, prevention, and recovery-support information.
- Designed the application to provide accessible responses to awareness-related questions, with English and Tamil knowledge support.
- Integrated an XGBoost prediction component and built the Streamlit-based web application.

DEPLOYED MACHINE LEARNING APPLICATIONS

Deployed selected ML projects as interactive web applications using Streamlit:

- Server Performance Anomaly Detection - One-Class SVM
- User Behavior Anomaly Detection - Isolation Forest
- Machine Sensor Anomaly Detection - Isolation Forest

ADDITIONAL MACHINE LEARNING EXPERIENCE

Regression	Linear Regression, Multiple Linear Regression, Polynomial Regression
Classification	Logistic Regression, KNN, Decision Tree, Random Forest, Naive Bayes, SVM
Ensemble Learning	Voting Classifier, Bagging, AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost
Unsupervised Learning	K-Means, Hierarchical Clustering, DBSCAN, OPTICS, GMM, PCA
Anomaly Detection	Z-Score, IQR, Isolation Forest, One-Class SVM
Model Optimization	GridSearchCV, RandomizedSearchCV, Cross Validation

DEVELOPMENT EXPERIENCE

- Applied Python fundamentals including data types, control flow, functions, collections, comprehensions, exception handling, file handling, and JSON.
- Practiced Object-Oriented Programming concepts: inheritance, polymorphism, aggregation, and composition.
- Building foundational backend development experience using Python, FastAPI, Pydantic, and Uvicorn, with REST API fundamentals.
- Practicing data structures and algorithmic problem solving in Python.
- Used Git and GitHub for version control and portfolio development.